Upcycling Bio-Based Precursors into Self-Supporting Carbon Electrodes through Surface Engineering for Supercapacitors

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Motivation

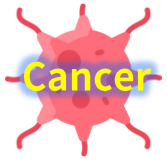




Rising CO_2, rising stress on our Earth



- PM2.5*
- CO gas
- PAHs*



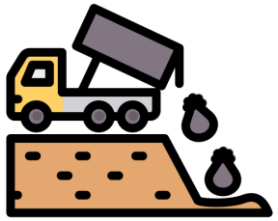
Bio-based waste represents an underutilized carbon resource.

*PM2.5: Particulate Matter 2.5 micrometers
*PAHs: Polycyclic Aromatic Hydrocarbons

Traditional waste management



Incineration (burn)



Landfill



Ocean dumping



Recycling



“Upcycling”

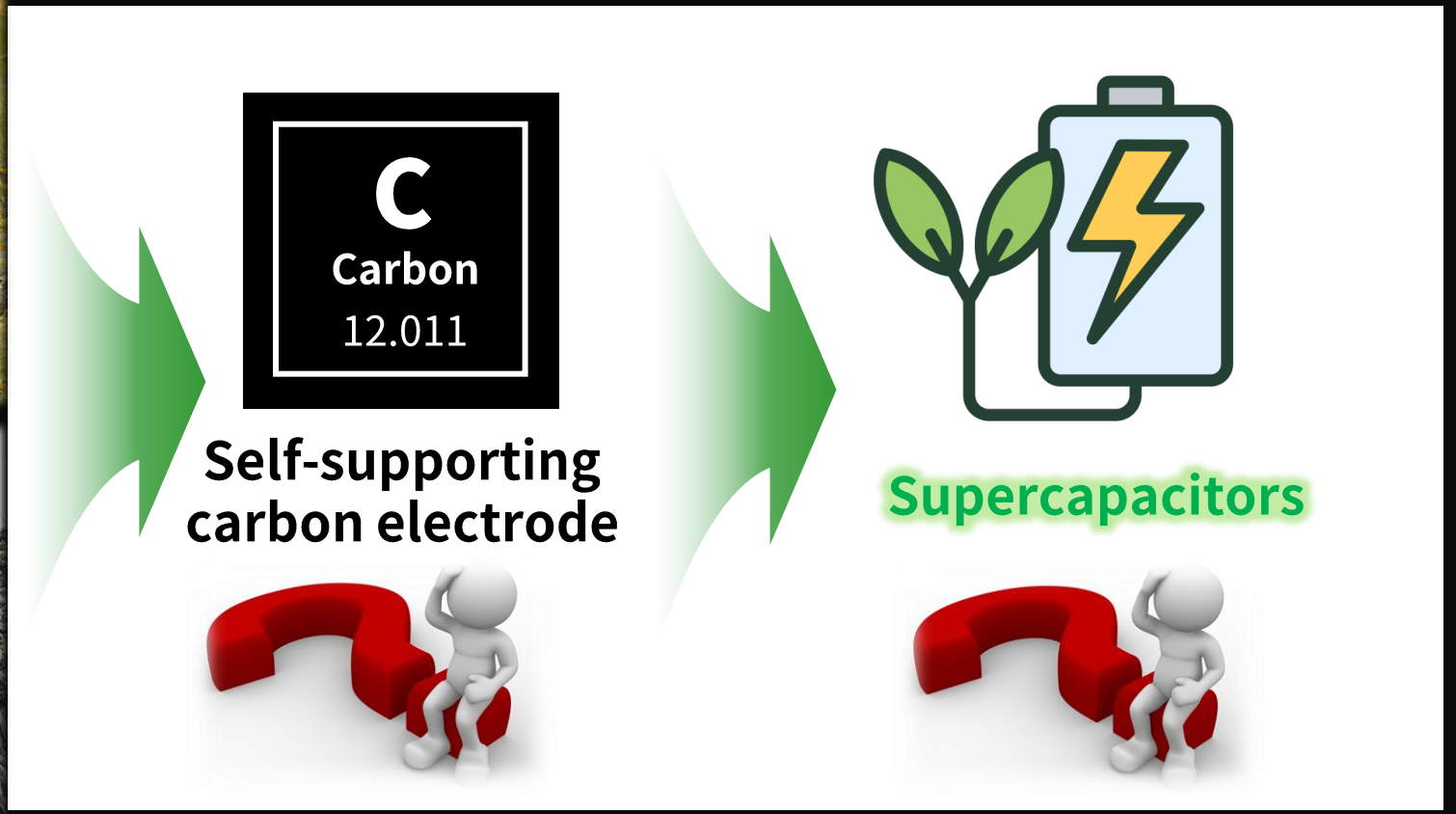


Upcycling is the process of **transforming waste materials** into new products with **higher value**.

Idea







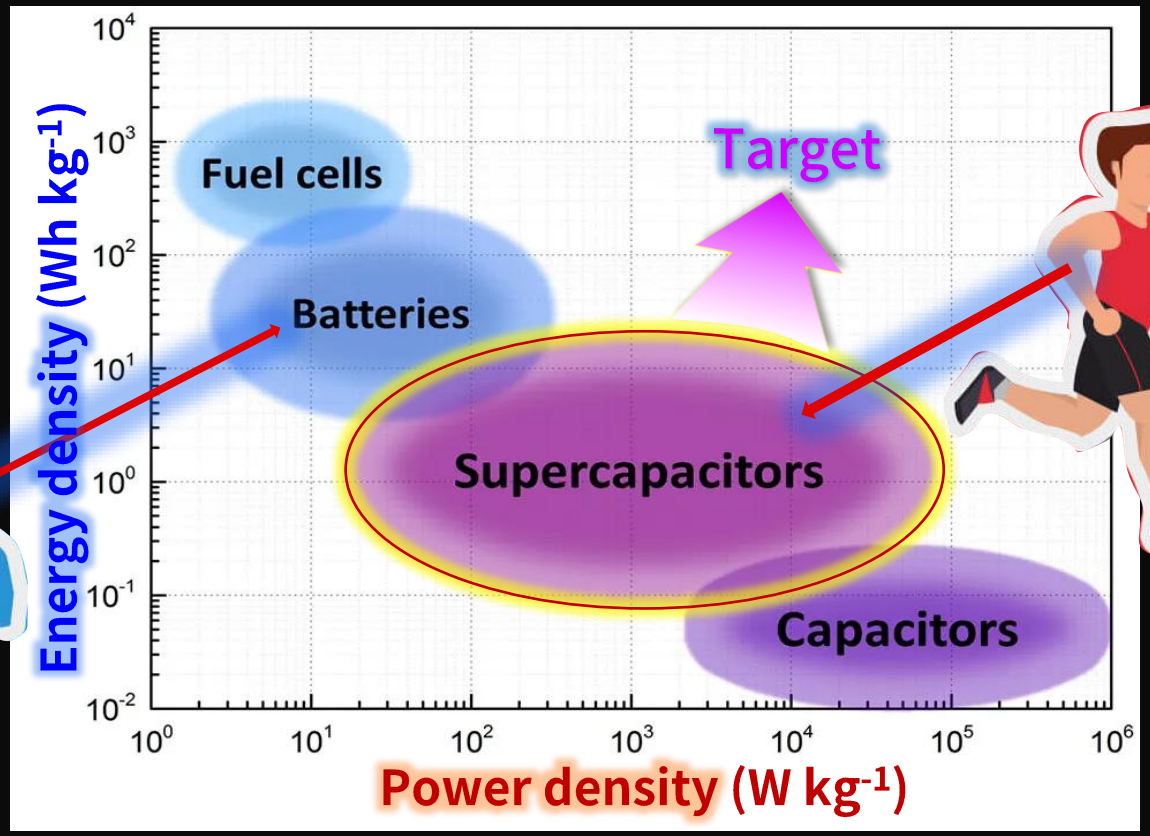
What if bio-waste could become an electrode directly for supercapacitors?

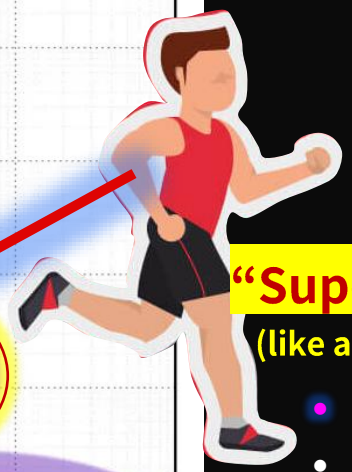
Why supercapacitors?



"Battery"
(like a sprinter)

- Fast and strong
- Poor cyclability
- Short life
- High energy





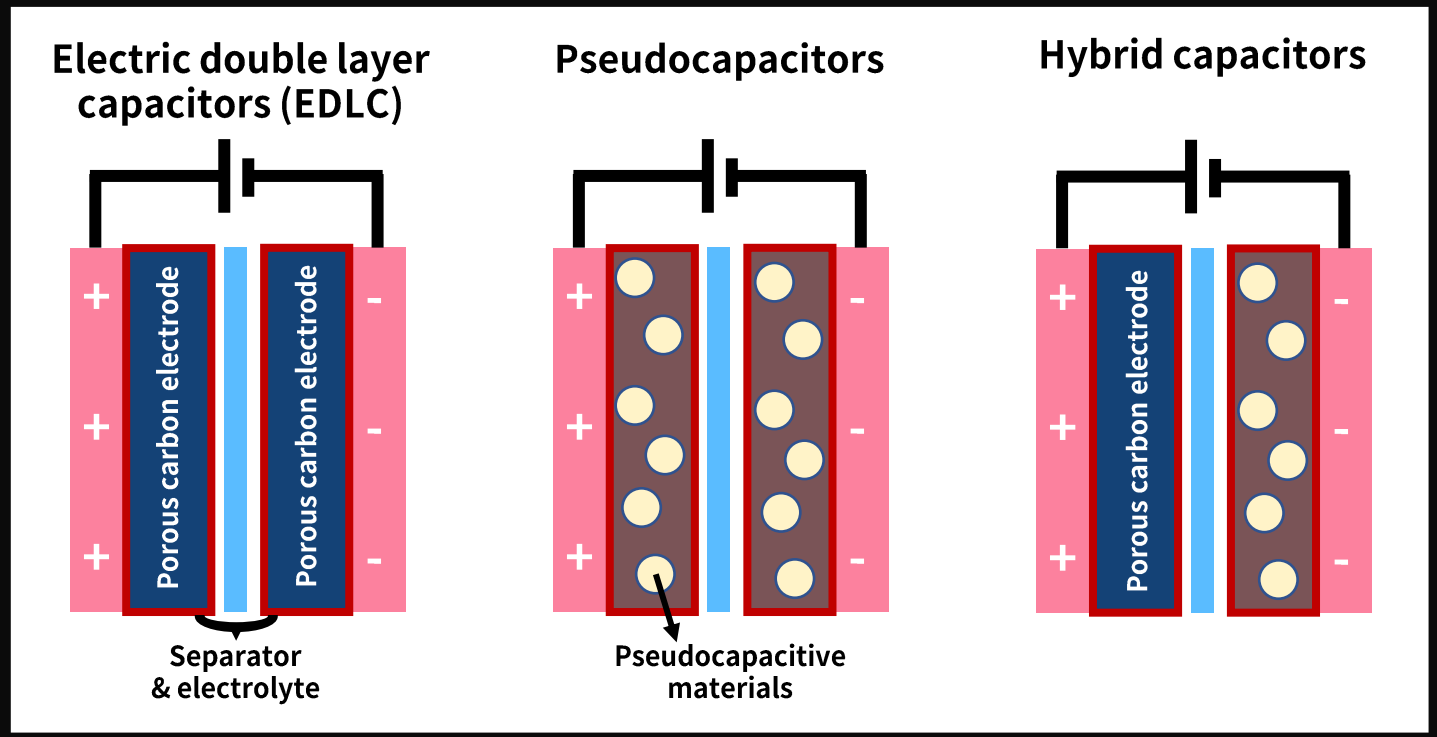
"Supercapacitor"
(like a marathon runner)

- **Long and enduring**
- Good cyclability
- Long life
- High power

Supercapacitors (SCs) are promising energy storage devices due to their **high power density** and **long cycle life**.

Supercapacitor Electrodes

❖ Supercapacitors are divided into three types:



What determines electrode performance?

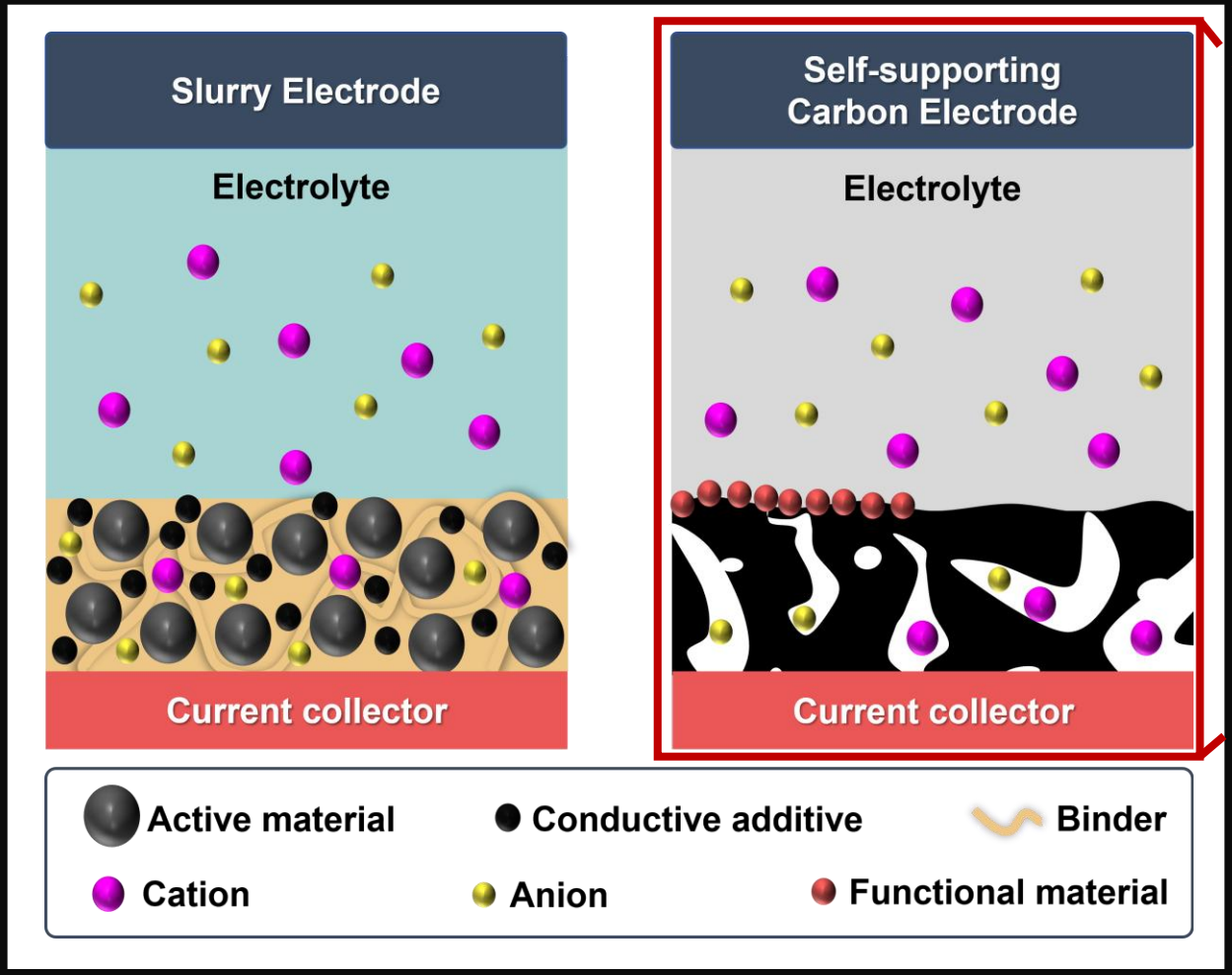
"Porosity engineering"

"Heteroatom doping"

"Redox-active material integration"

The key is to engineer the electrode—**not just the material.**

Why self-supporting carbon electrodes?



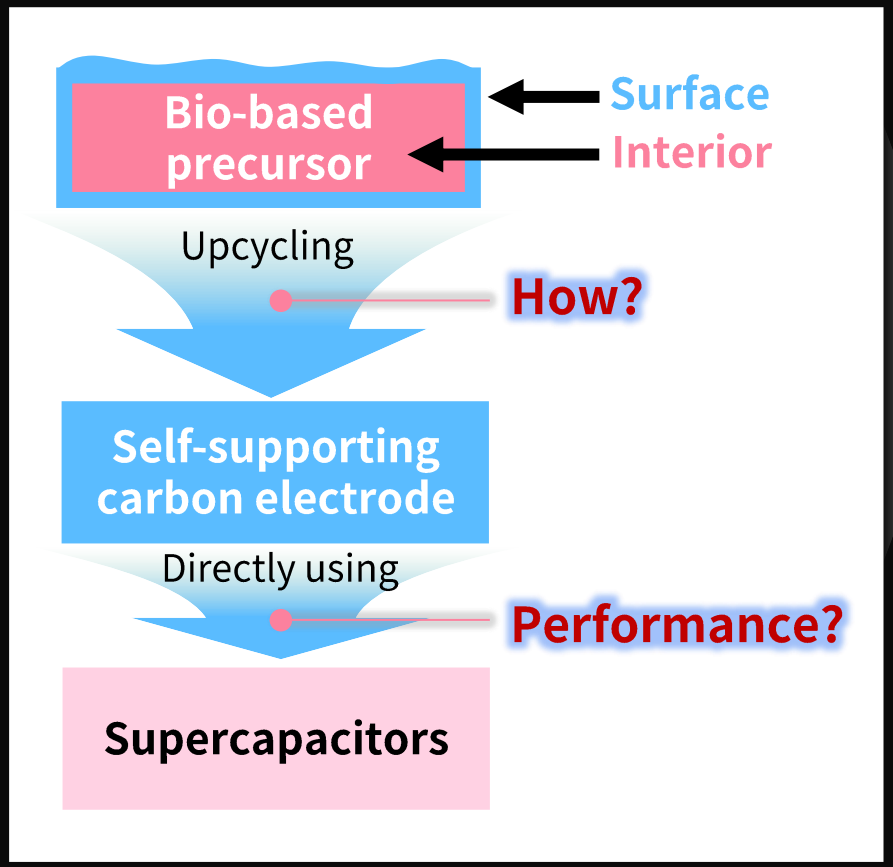
Self-supporting carbon electrode from bio-based precursors:

- Binder-free architecture
- Inherent carbon framwork
- Simply electrode fabrication

"A self-supporting advantages"

Why not turn bio-based precursors into the electrodes themselves?

Challenge



Bulk modification	Surface engineering
Use of harsh chemicals	Avoid harsh chemicals
Chemical waste generation	Environmental friendly
Potential damage to the carbon framework	Maintain self-support structure

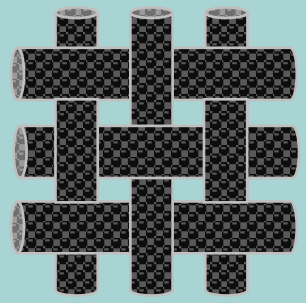
"Find the way to engineer the surface"

"How can surface engineering improve energy storage performance?"

Strategy

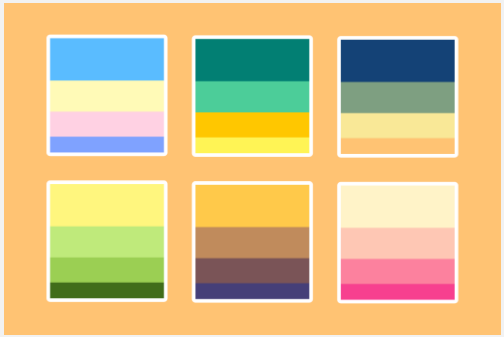
Self-Supporting Carbon Electrodes

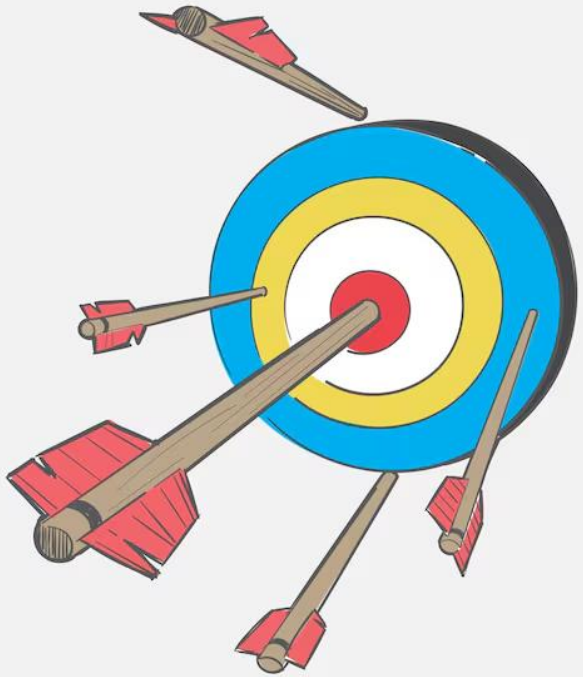
SURFACE ENGINEERING

Porosity engineering	Heteroatom* doping	Metal oxide integration
		
Study 1	**Study 2**	**Study 3**
Maple leaf	Cotton fabric	Cotton fabric

*Heteroatom: An organic molecule that is not carbon or hydrogen (O, N, S, P, Cl, etc.)

"To improve the energy storage performance of self-supporting carbon electrodes through surface engineering."



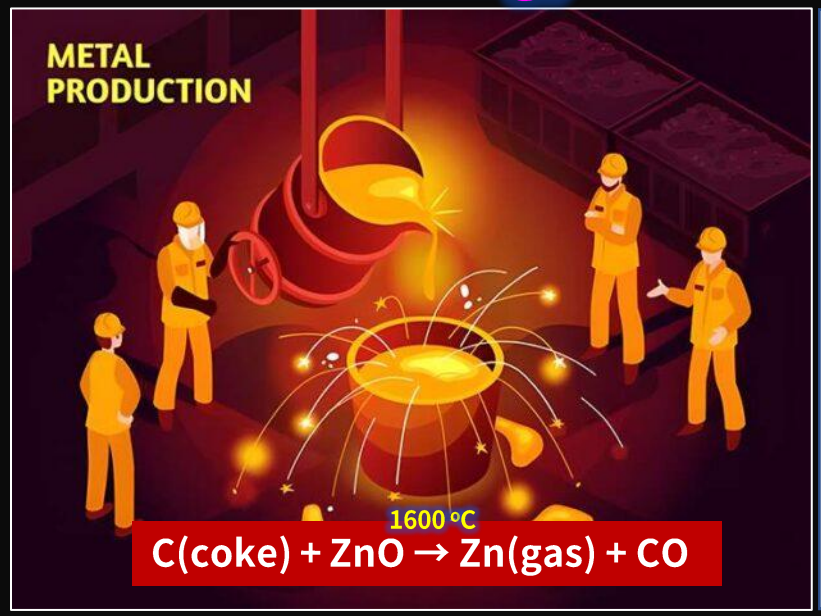


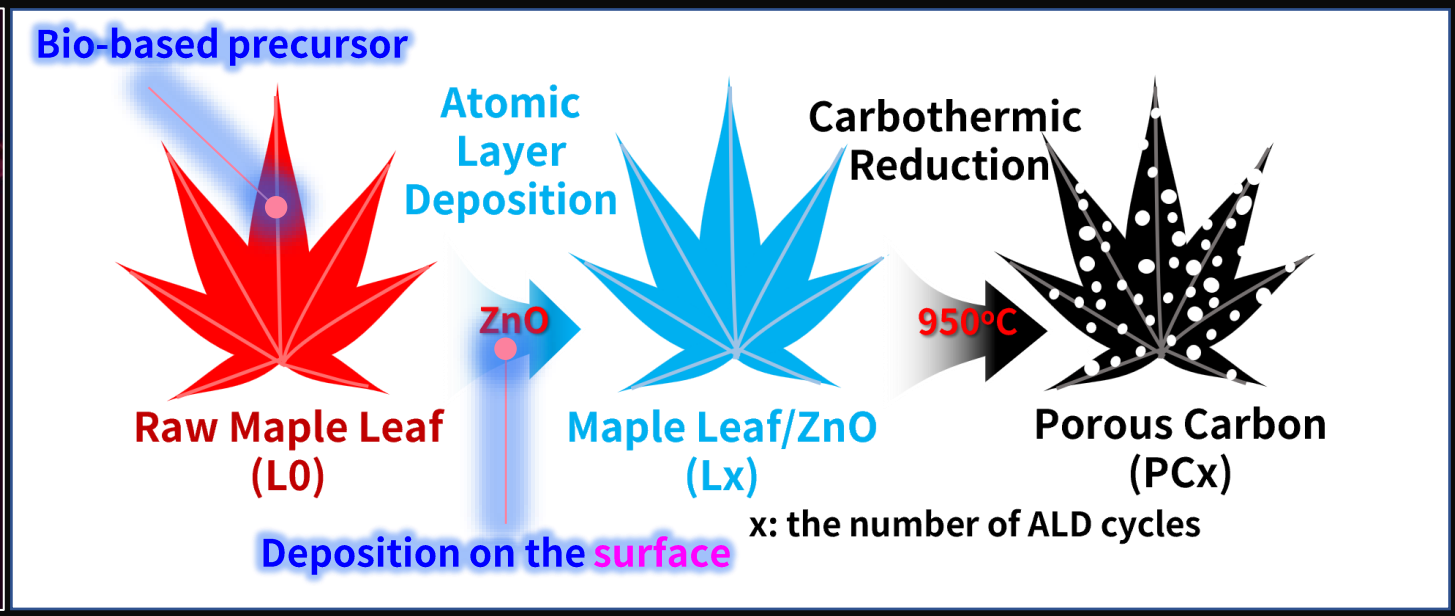
My first study—Porosity engineering

TOPIC 1

"How can porosity of leaf-derived carbon be engineered by ALD* deposition?"

Metallurgy



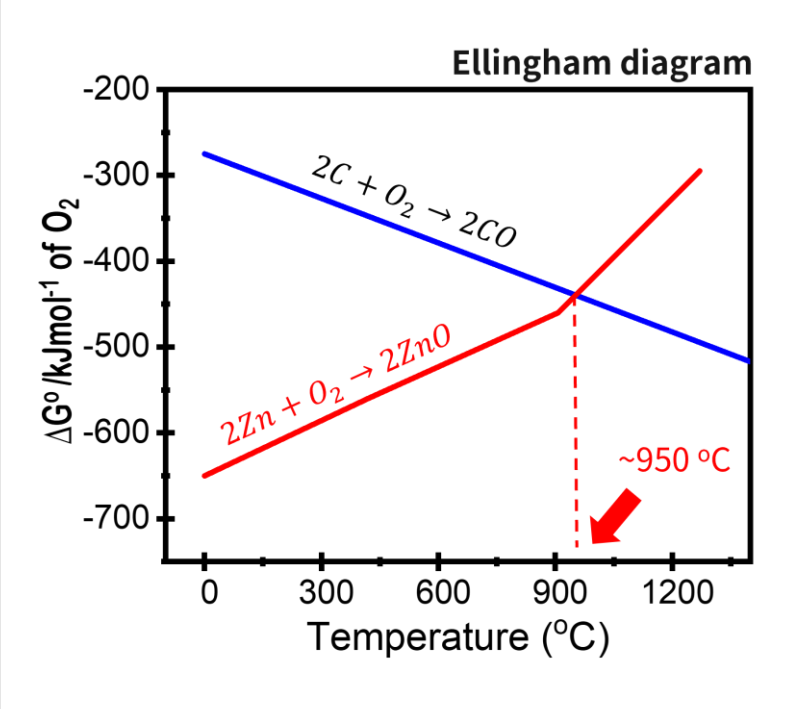


This strategy was inspired by metallurgy of Coke and Zinc Oxide but reserved.

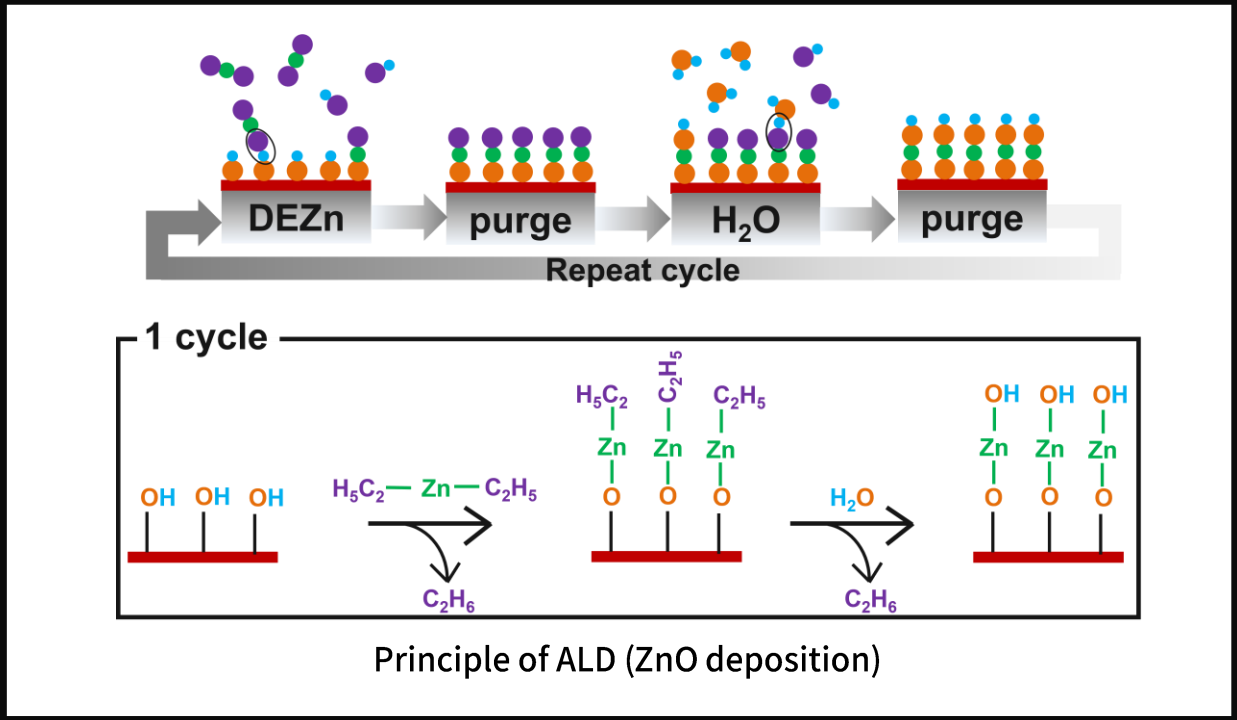
*ALD: Atomic Layer Deposition

My first study—Porosity engineering

"Why ZnO?"



"Why ALD*"?



Principle of ALD (ZnO deposition)

Carbothermic reduction

$$C\ (leaf) + ZnO\ (s) \xrightarrow{\sim 950\ ^{o}C} C\ (porous\ carbon) + Zn(g) + CO(g)$$

($T_{boil\,(Zn)}$ = 907 °C)

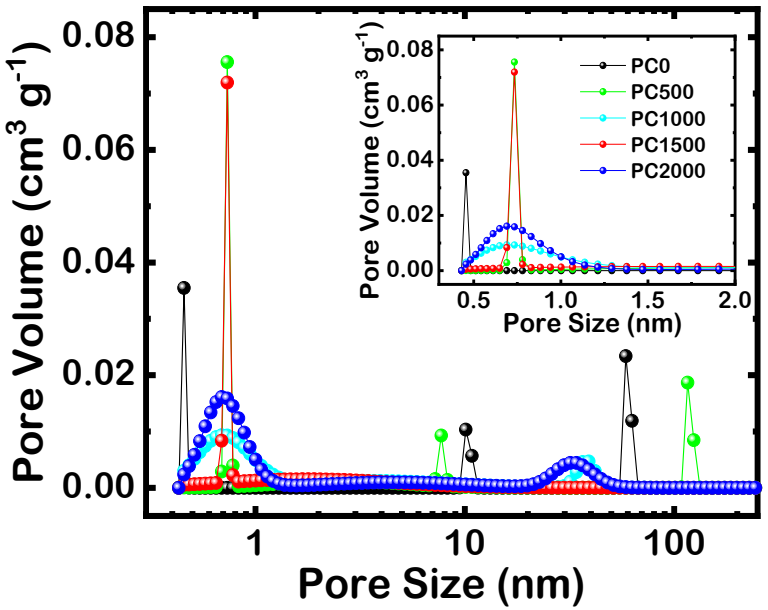
"Uniform deposition"

"Controlled coating thickness"

*ALD: Atomic Layer Deposition

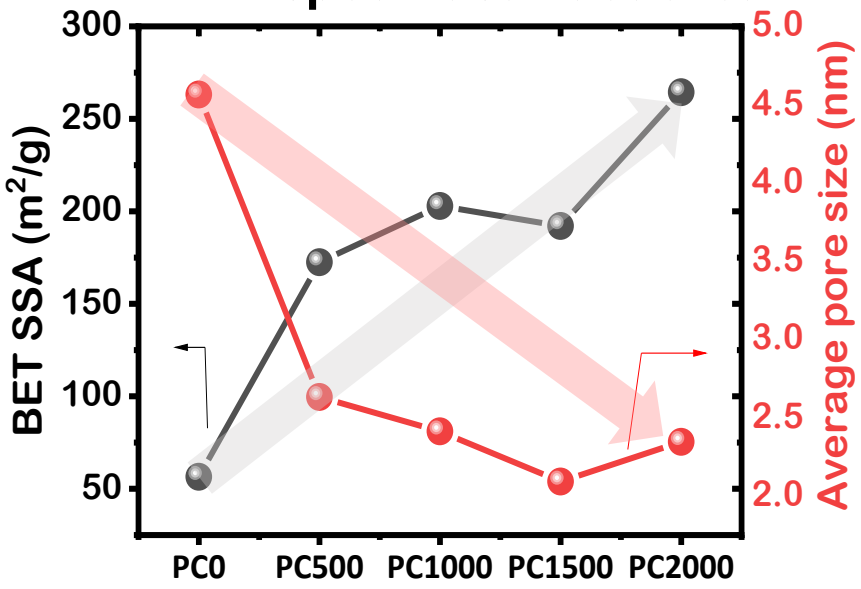
My first study—Porosity engineering

❖ Pore size distribution



Increase of micropores

❖ BET specific surface area

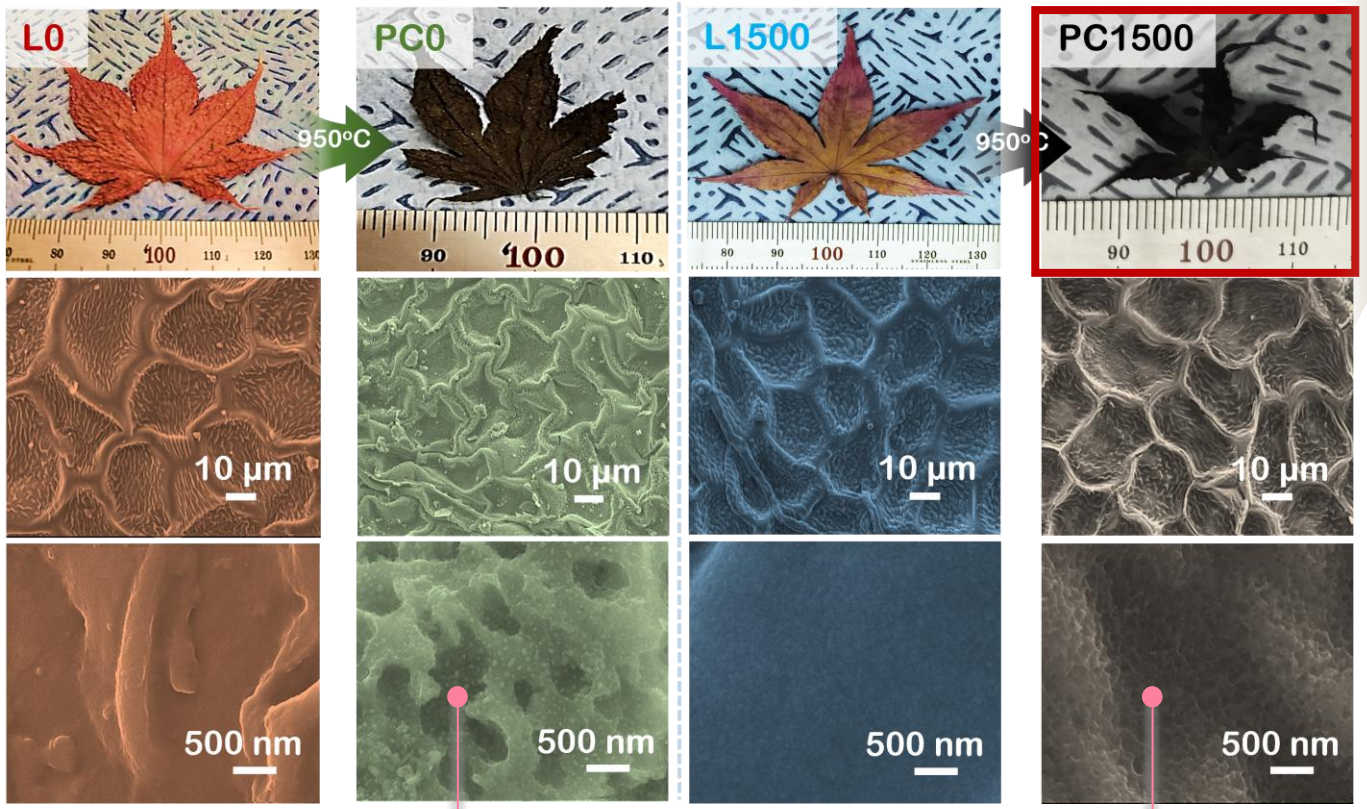


Increase of surface area

The carbothermic reduction of ZnO and the accompanied solid-state etching successfully occur.

My first study—Porosity engineering

❖ Morphology



Larger pores

Smaller pores

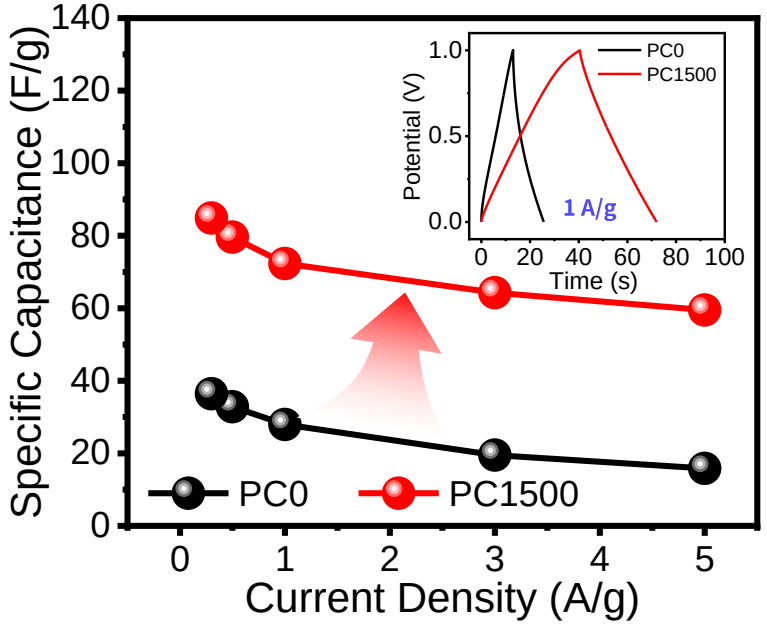
Binder-free carbon electrode for EDLC supercapacitors

Energy storage performance?

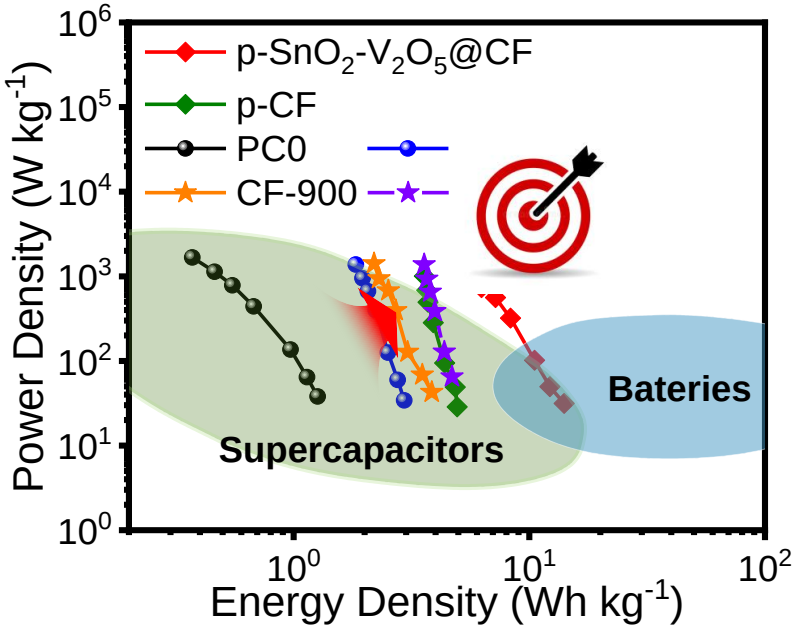


My first study—Porosity engineering

❖ Specific capacitance



❖ Ragone plot



PC1500 has highest energy storage performance

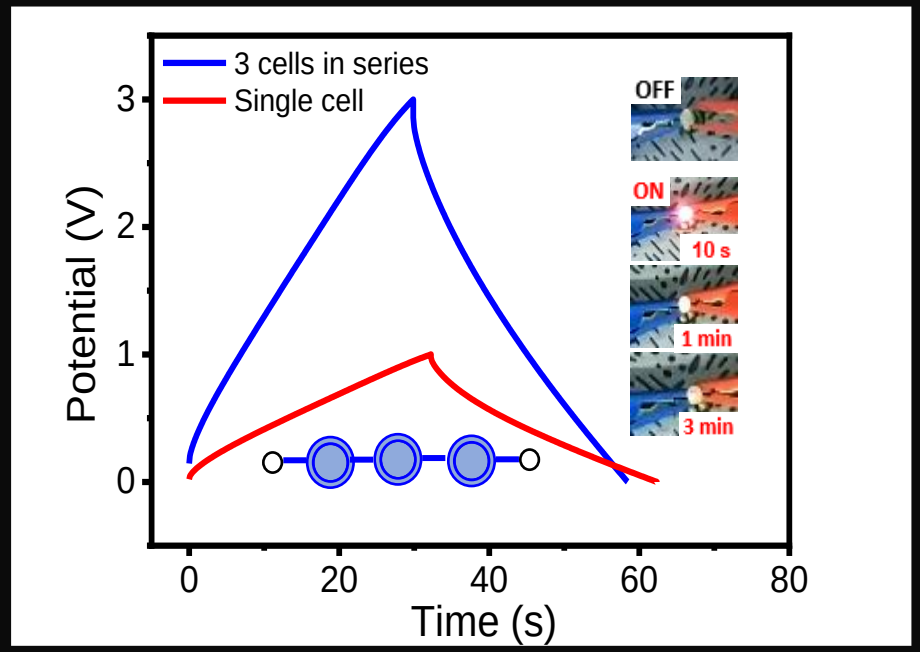
→ Highlighting the importance of pore-structure optimization.

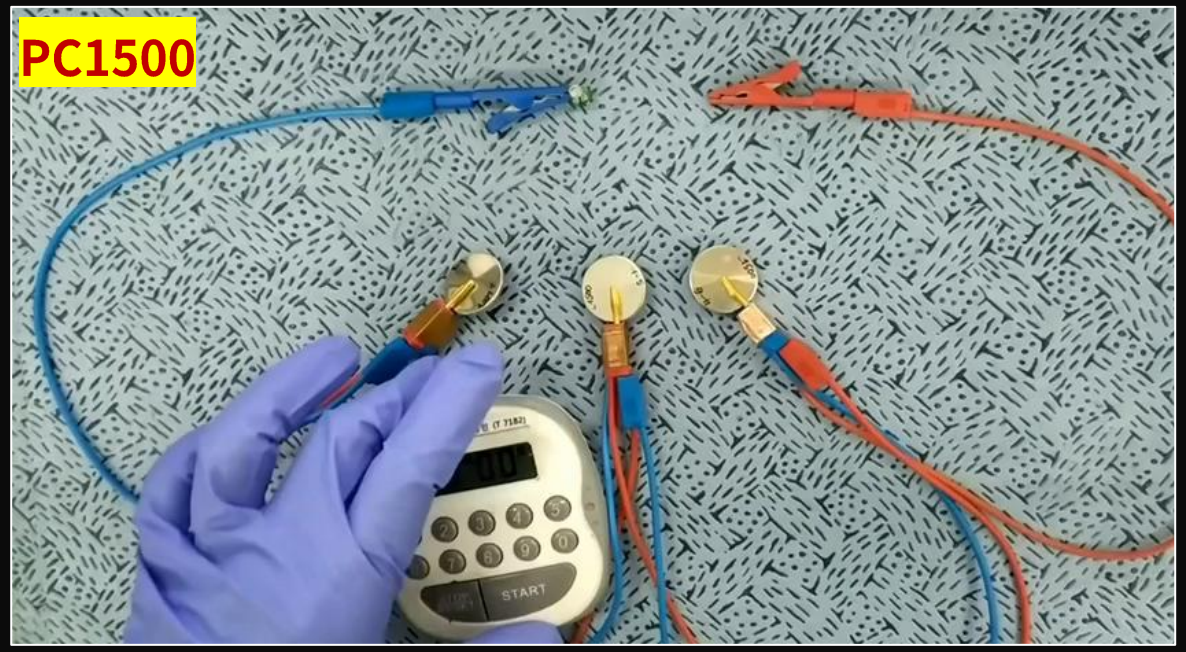
Energy density of PC1500 has increased

Proposed target

My first study—Porosity engineering

Lighten LED

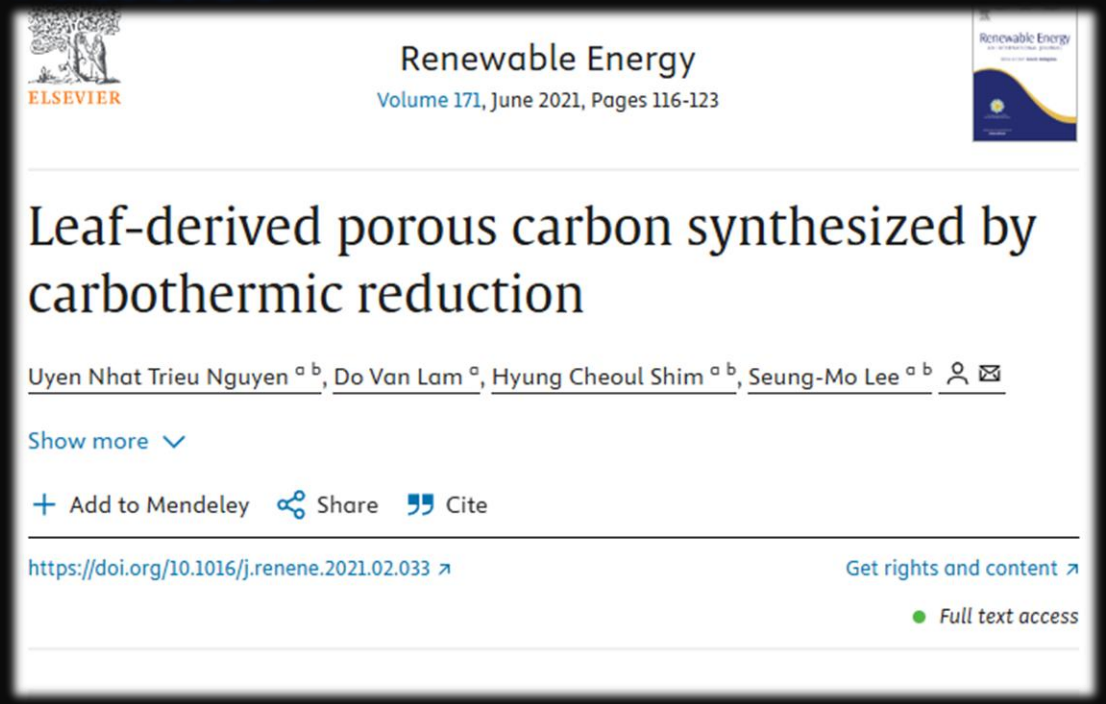




The three-coin cells in series were able to lighten a red light-emitting diode for over 3 minutes.

My first study—Porosity engineering

Publication



ELSEVIER

Renewable Energy

Volume 171, June 2021, Pages 116-123

Leaf-derived porous carbon synthesized by carbothermic reduction

Uyen Nhat Trieu Nguyen [a b], Do Van Lam [a], Hyung Cheoul Shim [a b], Seung-Mo Lee [a b]

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https://doi.org/10.1016/j.renene.2021.02.033

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Remaining challenge

The energy density is still within the typical range of conventional supercapacitors.

Further improvement!

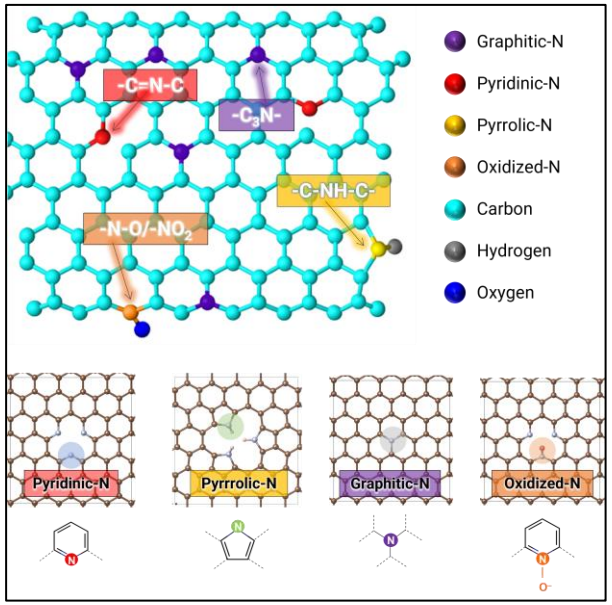
Next direction:

Further surface engineering through heteroatom doping for another bio-based precursor should be studied.

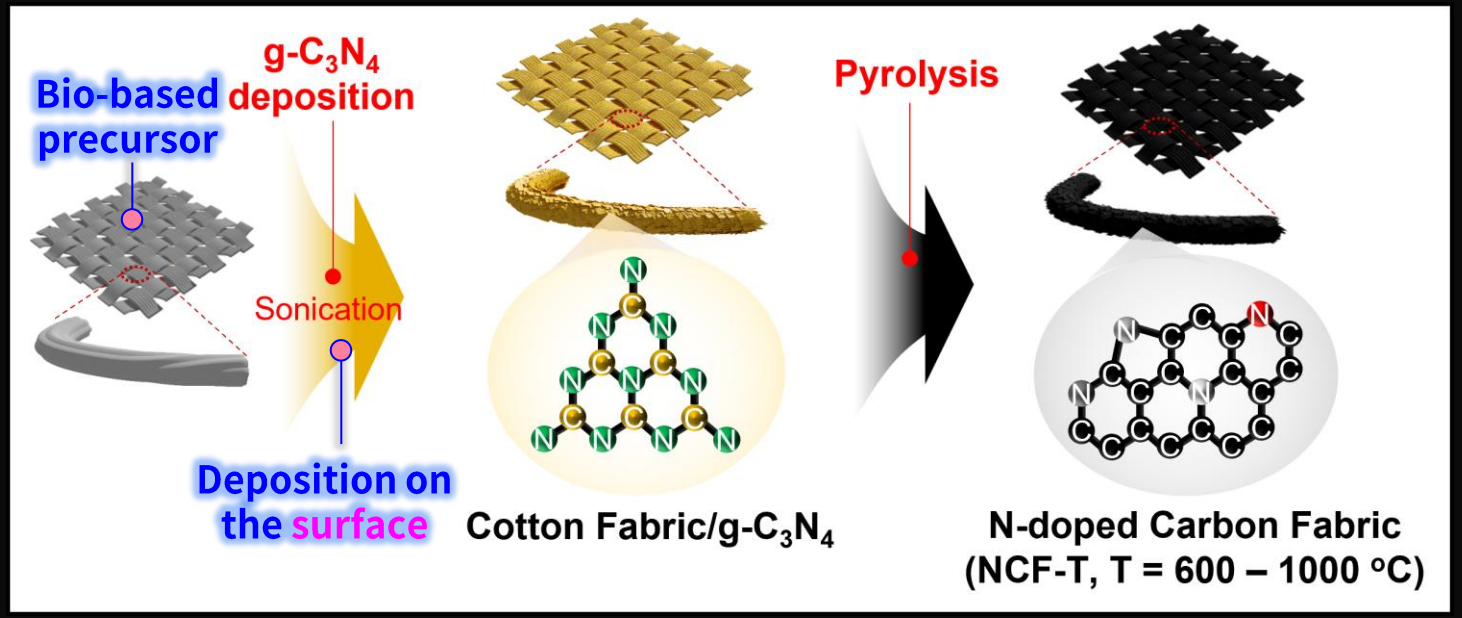
My second study—Heteroatom doping

TOPIC 2

4 types of Nitrogen doping



"How can control Nitrogen doping of cotton fabric-derived carbon?"

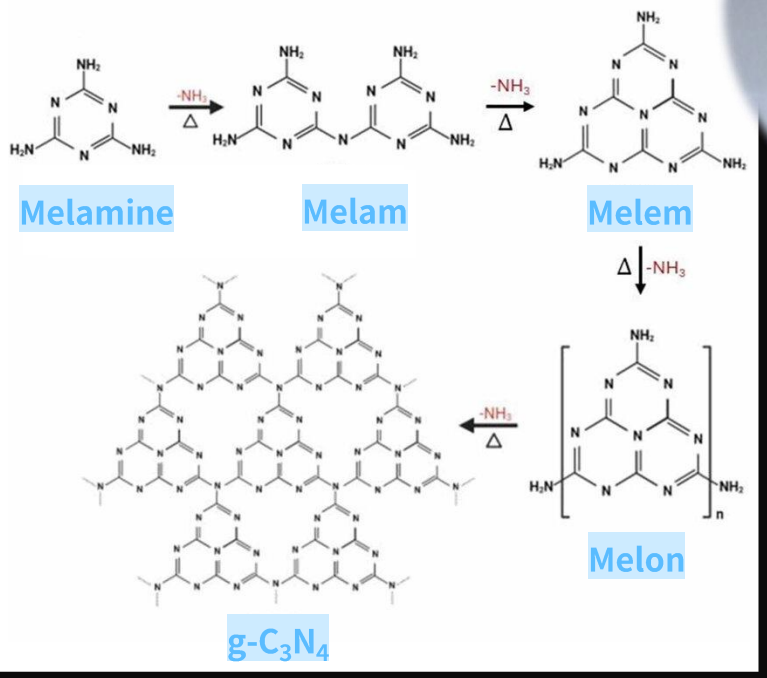


Inspired by g-C_3N_4, a surface engineering strategy was developed for nitrogen doping of carbon.

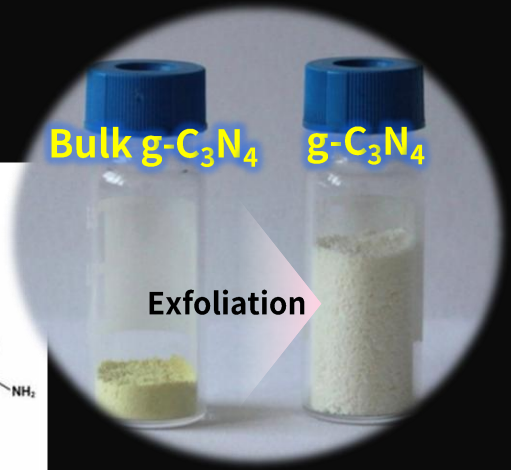
My second study—Heteroatom doping

"What is g-C_3N_4?"

❖ Conventional synthesis of g-C_3N_4



Materials Research Express (2026), 13, 125001.



Scientific reports (2016), 6, 28558.

"Why g-C_3N_4 & Why sonication?"

- ✓ Nitrogen-rich structure
- ✓ Nanolayer coating on cotton fabric

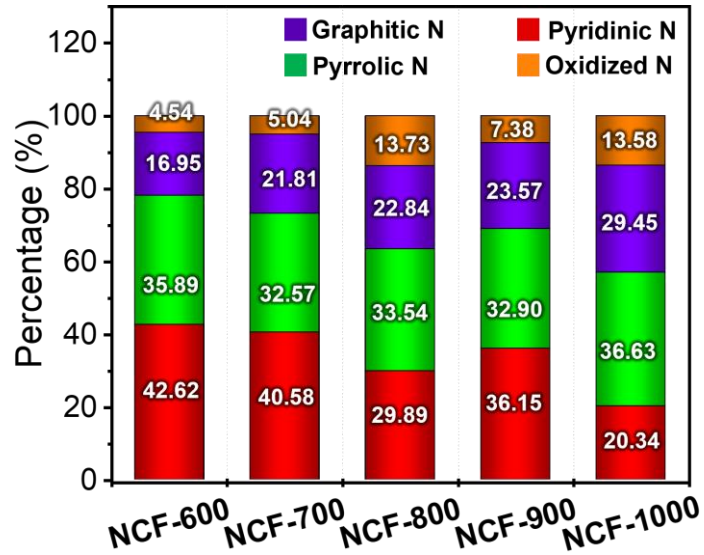
Our strategy

- ✓ Cotton fabric immersion in ethanol
- ✓ Direct sonication
- ✓ In situ exfoliation of g-C_3N_4

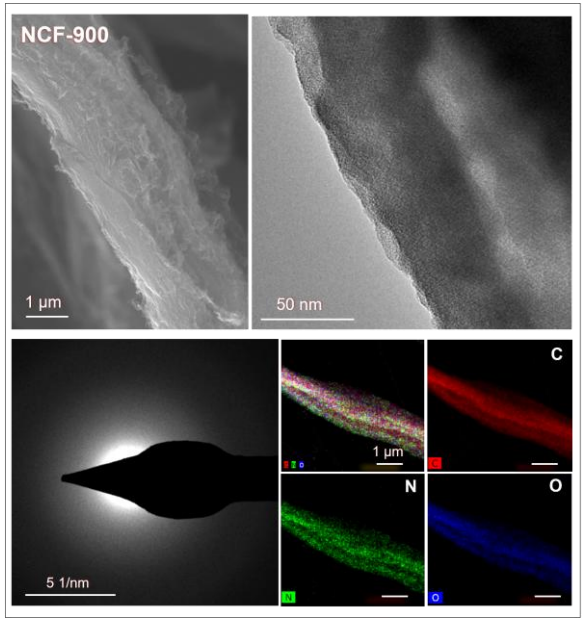
This strategy is expected to promote the formation of an N-doped nanostructure on carbon fabric after pyrolysis.

My second study—Heteroatom doping

❖ Relative proportions of nitrogen configurations



❖ SEM, TEM, and SAED



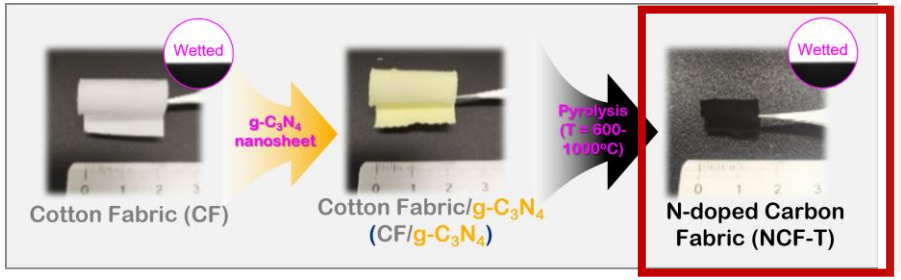
Effect of Pyrolysis Temperature on Nitrogen Configurations

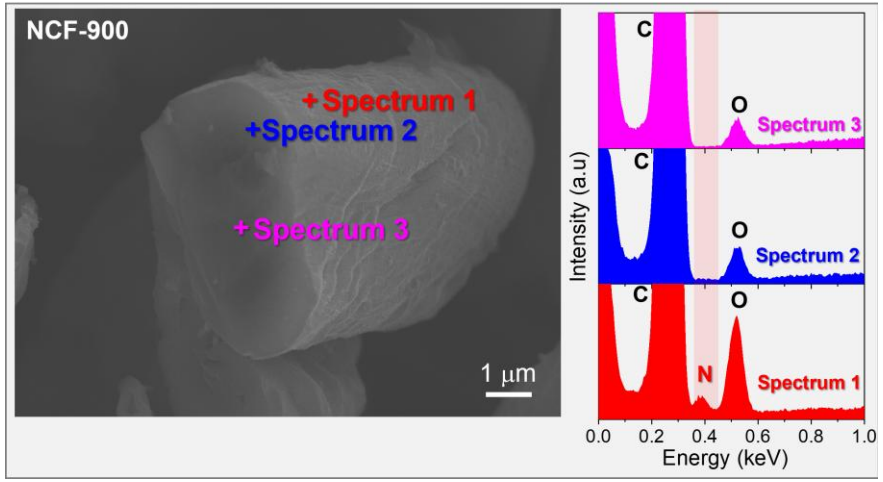
Nitrogen Doping and Nanofilm Formation on Carbon Fibers

Nitrogen-doped nanostructures were successfully formed on carbon fabric.

My second study—Heteroatom doping

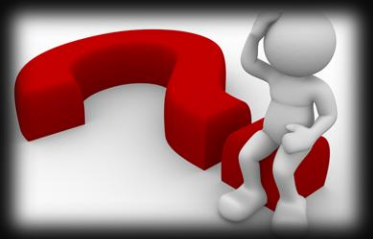
❖ Morphology





Self-supporting carbon electrode for both EDLC* & Pseudocapacitor

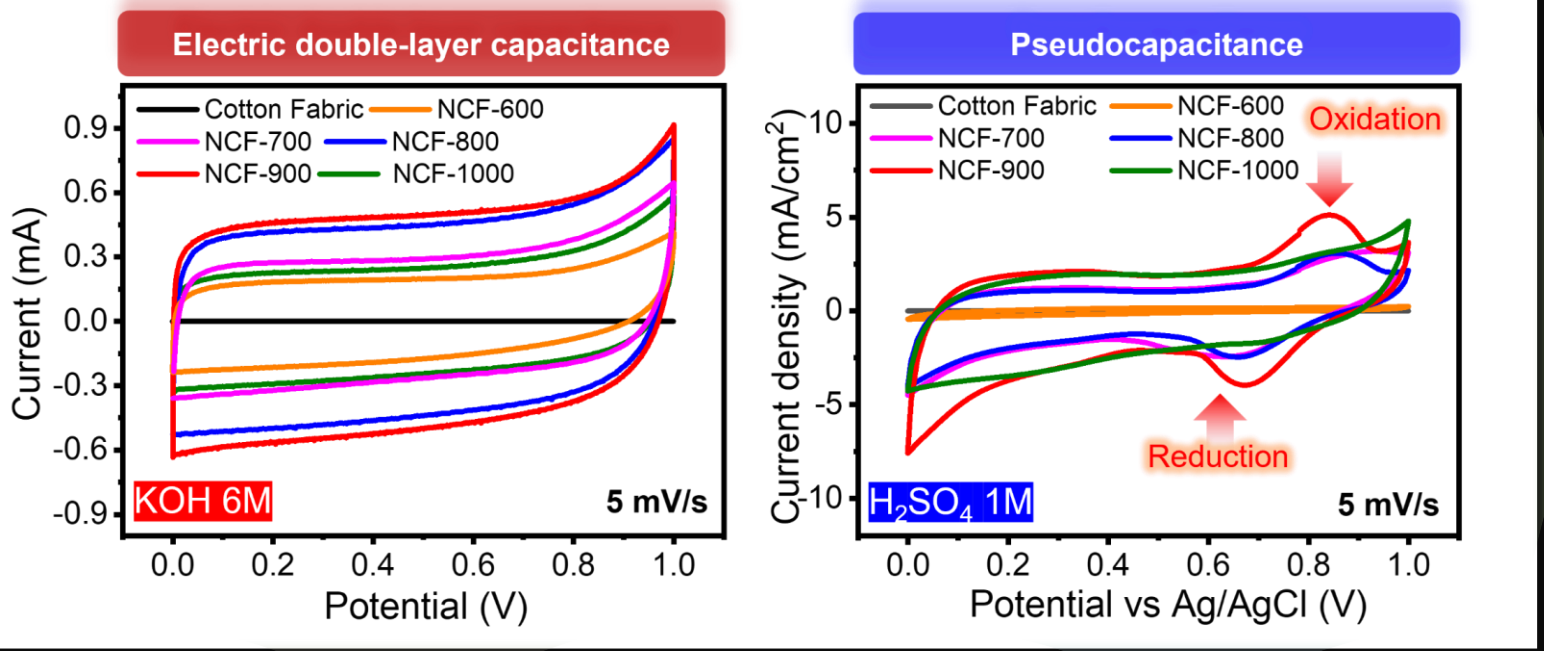
Energy storage performance?



*EDLC: Electric Double Layer Capacitors

My second study—Heteroatom doping

❖CV* curves



NCF900 has highest energy storage performance

Rectangular profiles

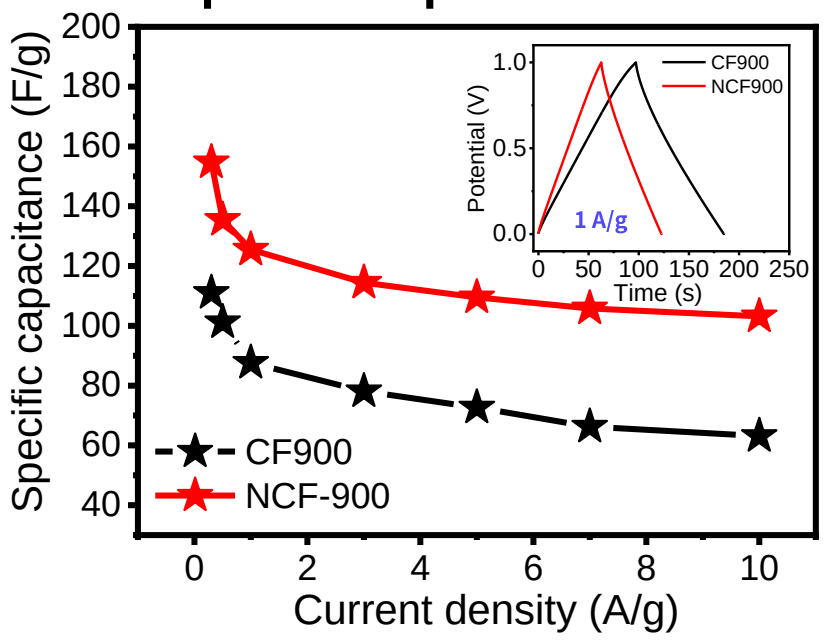
Broader redox features

Nitrogen doping enhances pseudocapacitive behavior through reversible redox reactions.

*CV: Cyclic Voltammogram

My second study—Heteroatom doping

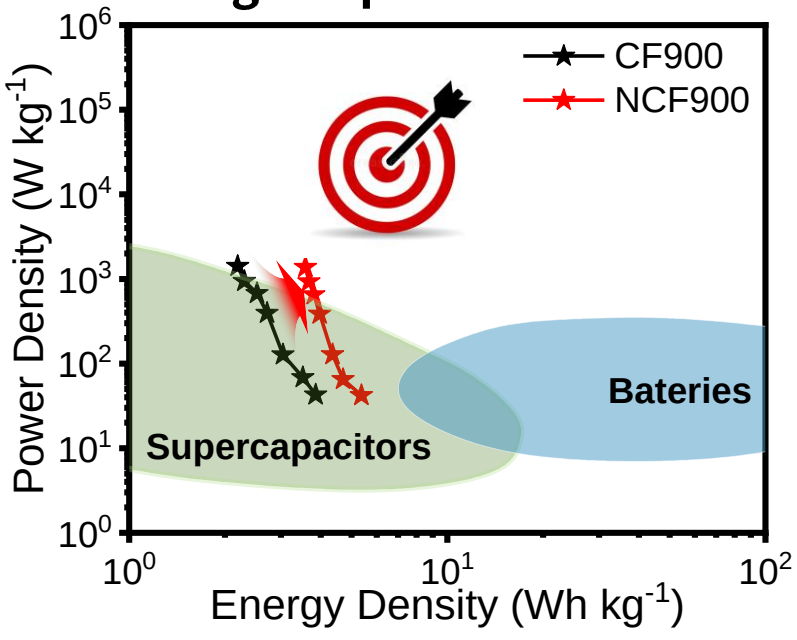
❖ Specific capacitance



Higher specific capacitance across current densities

→ Improved charge-storage capability

❖ Ragone plot

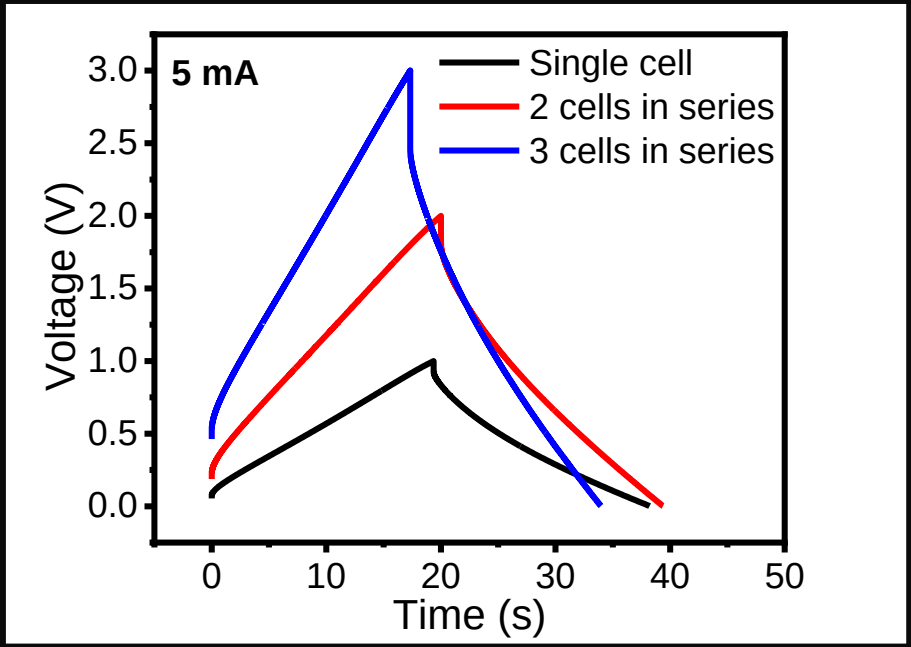


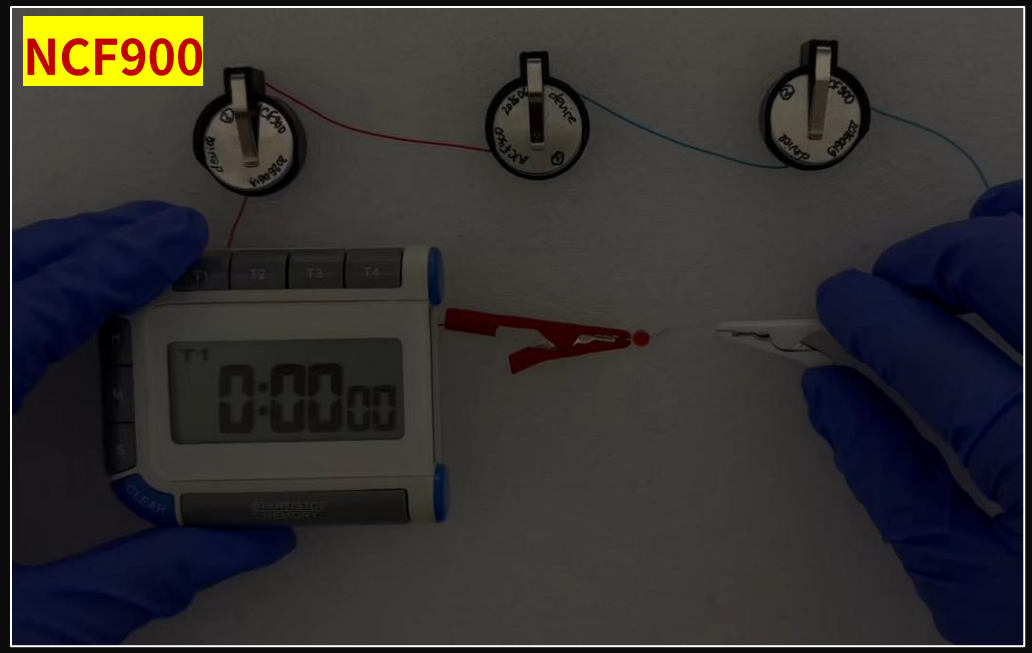
Energy density of NCF900 has increased

Proposed target

My second study—Heteroatom doping

Lighten LED





Cotton fabric is a promising bio-based precursor for self-supporting carbon electrodes.

The three-coin cells in series were able to lighten a red light-emitting diode for over 7 minutes.

My second study—Heteroatom doping

Publication



Applied Surface Science 747 (2026) 167701

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Role of pyridinic nitrogen in redox activity of nitrogen-doped carbon fabrics

Uyen Nhat Trieu Nguyen [a,b], Moonwon Lee [c,d], Kanghoon Yim [c], Dae Hoon Lee [a], Seung-Mo Lee [a,b,*]

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[b] University of Science and Technology (UST), 217 Gajeong-ro, Yuseong-gu, Daejeon 34113, South Korea
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[d] Graduate School of Energy Science and Technology (GEST), Chungnam National University, 99 Daehak-ro, Yuseong-gu, Daejeon 34134, South Korea

Remaining challenge

Energy density has not yet increased significantly.

Further improvement!

Next direction:

Further surface engineering through metal oxide integration for cotton fabric should be studied.

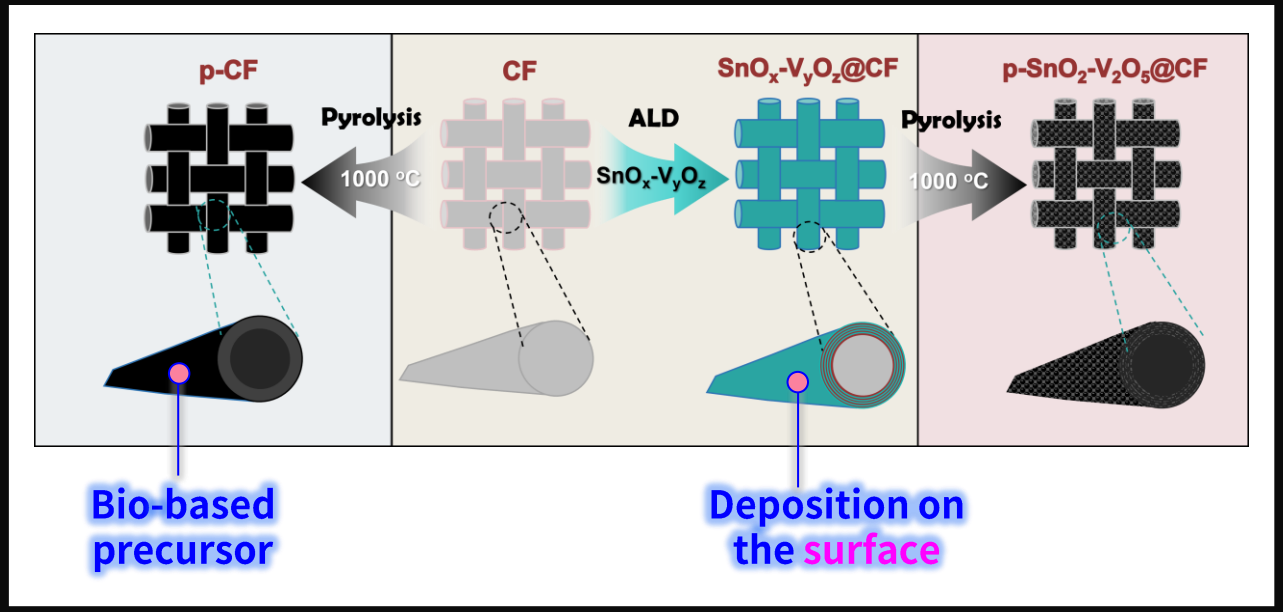
My third study—Metal integration

TOPIC 3

"How can metal oxides be integrated into cotton fabric-derived carbon through ALD?"

Why metal oxide integration?

- ✓ **Carbon alone:**
  Limited performance
- ✓ **Metal oxide integration:**
  Enhanced performance

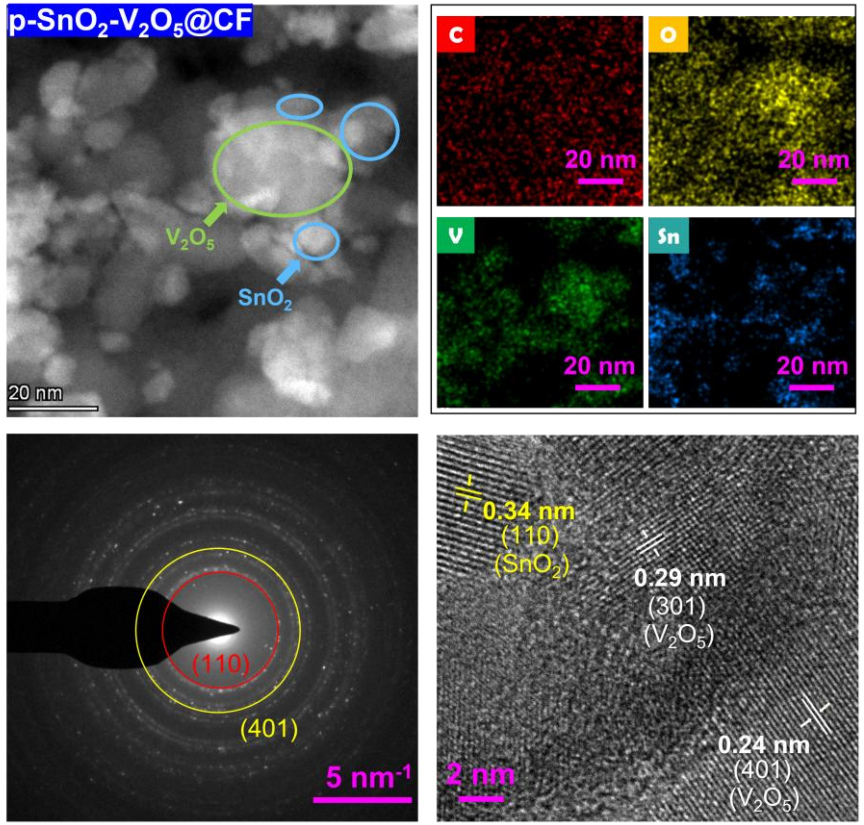


Metal oxide integration was studied as the next surface engineering strategy—by using ALD for deposition.

*ALD: Atomic Layer Deposition

My third study—Metal integration

❖ Characterization



Confirmed coexistence of SnO_2 and V_2O_5 phases

Successful integration of SnO_2/V_2O_5 onto the carbon fabric through ALD and pyrolysis.

*HR-TEM: High-Resolition Transmission Electron Microscopy

**SAED: Selected Area Electron Diffraction

My third study—Metal integration

editting

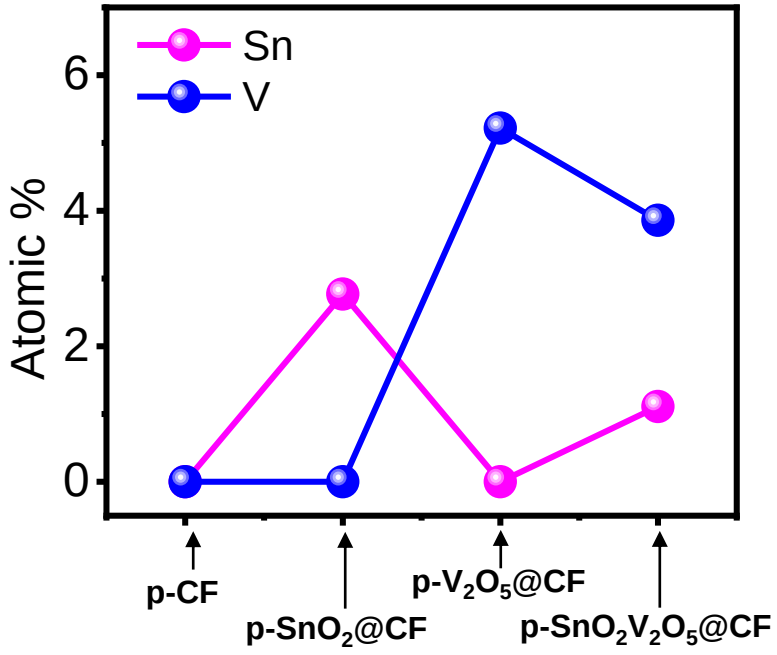
Real photo

"Why SnO_2?"

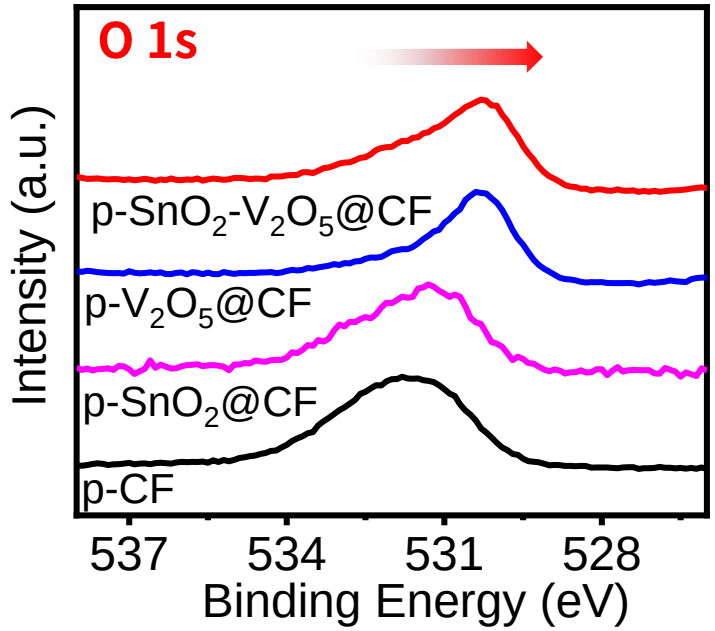
"Why V_2O_5?"

My third study—Metal integration

❖ Atomic percent of Sn and V



❖ O 1s XPS spectra

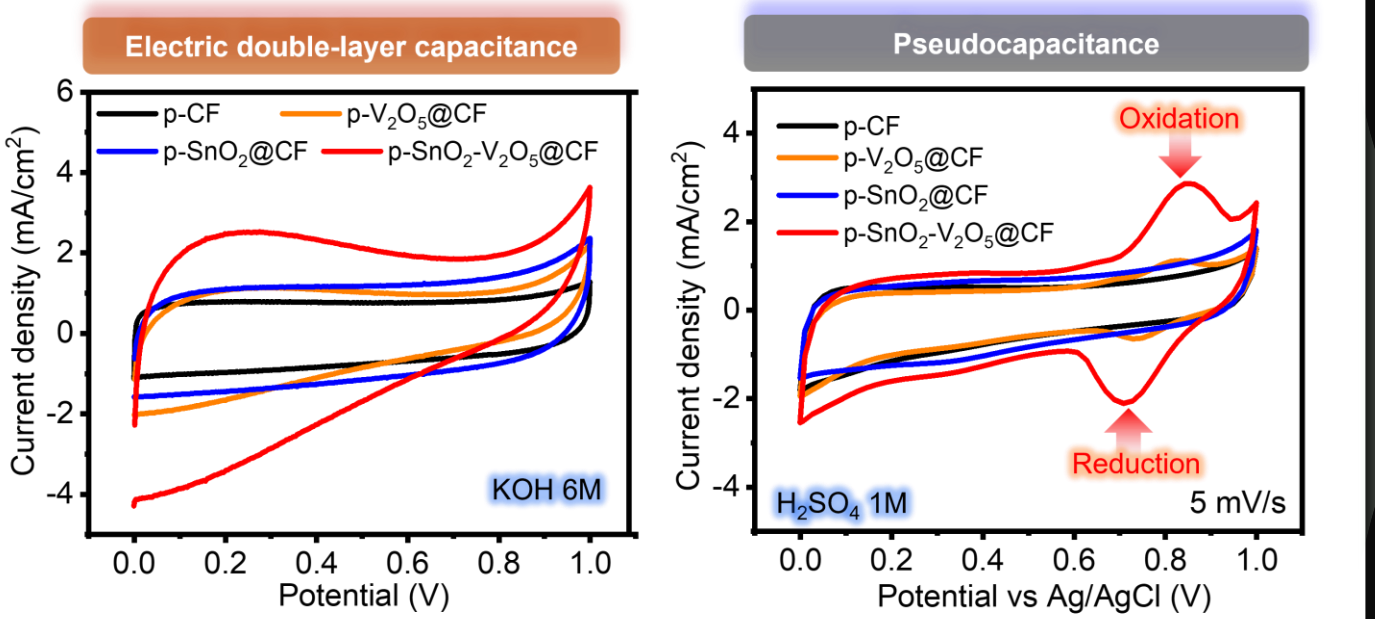


- Successful Sn and V incorporation
- Composition changes with oxide deposition
- Coexistence of Sn and V in the hybrid sample

*XPS: X-ray photoelectron spectroscopy

My third study—Metal integration

❖CV* curves



- ✓ V_2O_5 introduces distinct redox features.
- ✓ SnO_2–V_2O_5 integration produces the strongest redox response.

Non-rectangular CV profiles → Non-ideal EDLC behavior

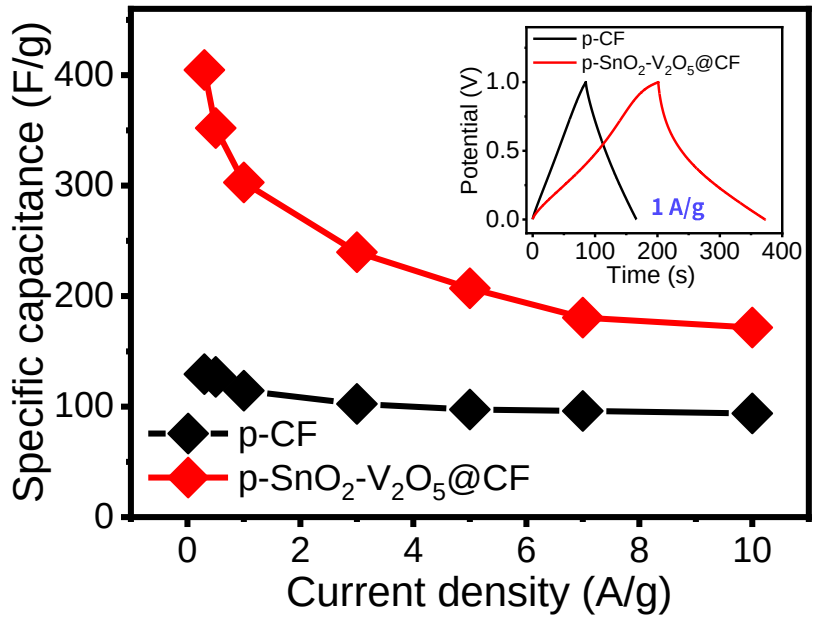
Redox peaks emerge in V_2O_5-containing electrodes.

SnO_2/V_2O_5 integration combines electric double-layer and redox contributions to enhance charge storage.

*CV: Cyclic Voltammogram

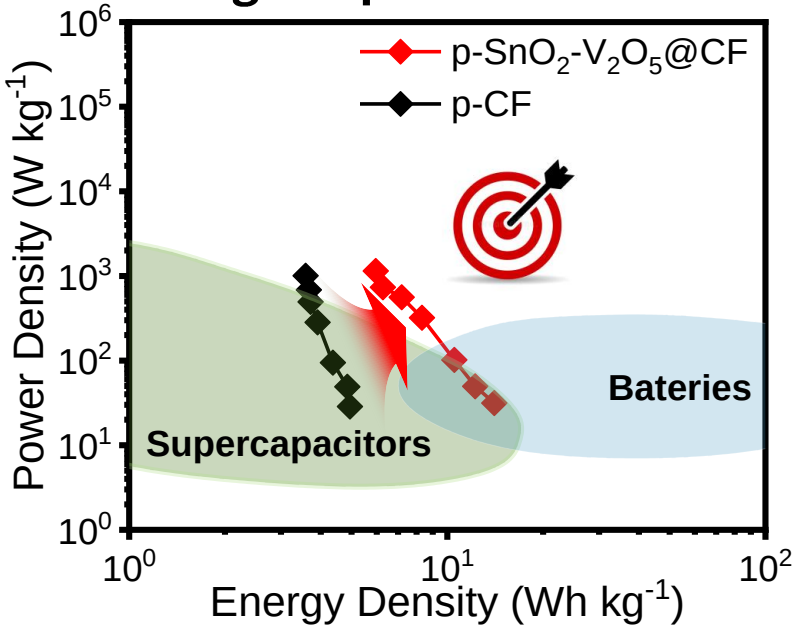
My third study—Metal integration

❖ Specific capacitance



Specific capacitance increased by ~2.8fold

❖ Ragone plot



Energy density of p-SnO_2–V_2O_5@CF has increased significantly

Proposed target

*ALD: Atomic Layer Deposition

My third study—Metal integration

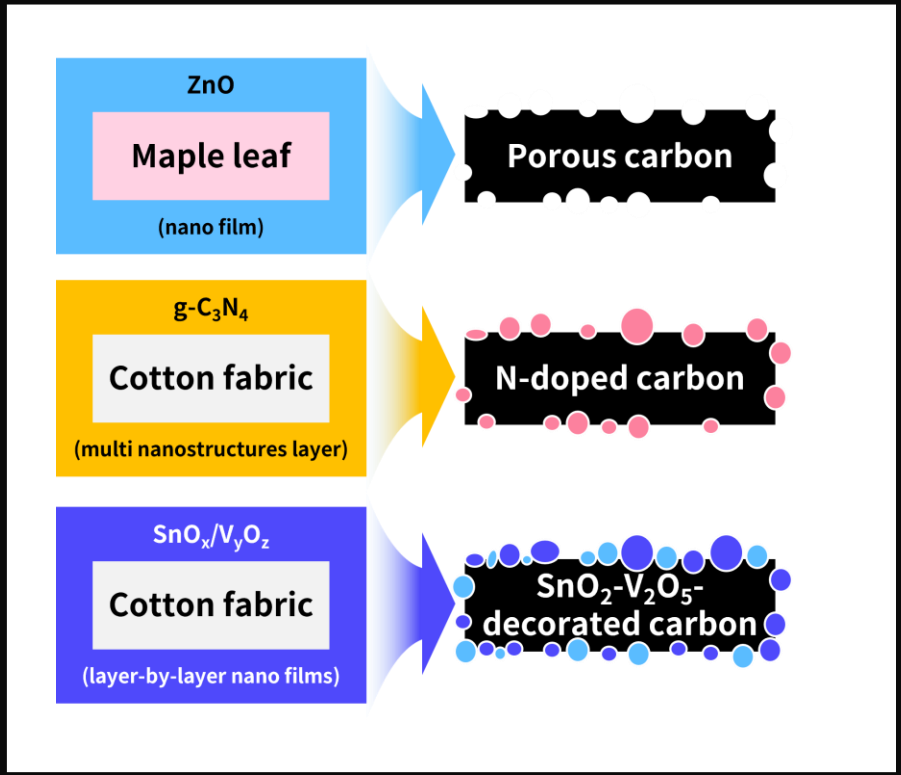
editting

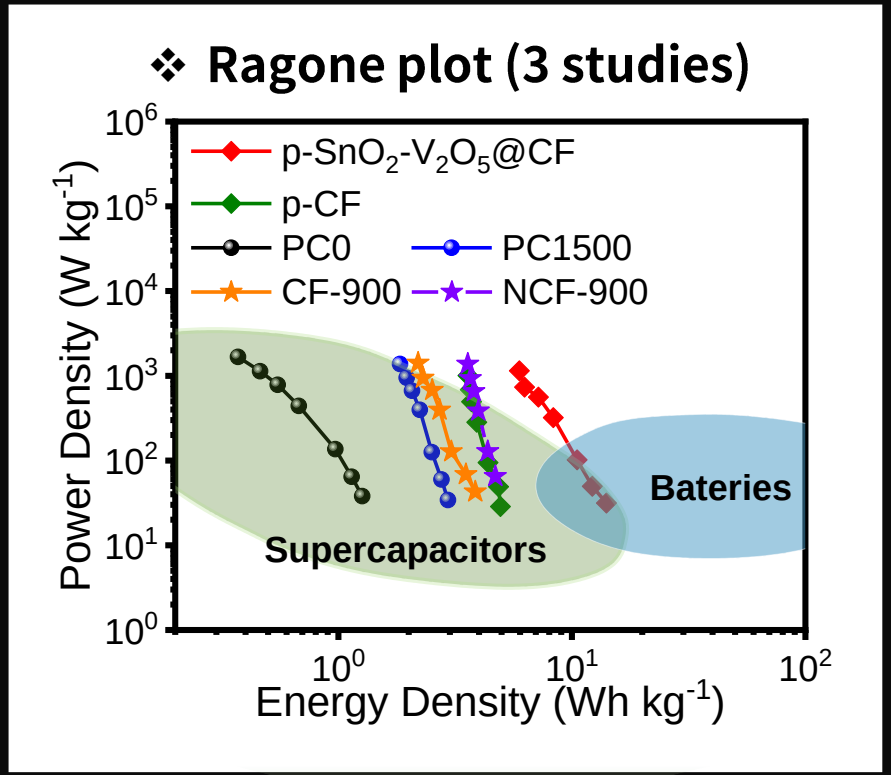
Lighten LED

I will do now.

Summary

editting





Future plan

editting

THANK YOU SO MUCH